

Analytical

Hough circle transform

①

Given,

edge points, $P_1 = (5, 4)$

$P_2 = (7, 6)$

Candidate centre, $C_1 = (2, 4)$

$C_2 = (4, 3)$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

for $C_1 = (2, 4)$:

for P_1 :

$$d = \sqrt{(5-2)^2 + (4-4)^2}$$

$$d = \sqrt{9} = 3$$

for P_2 :

$$d = \sqrt{(7-2)^2 + (6-4)^2}$$

$$= \sqrt{(5)^2 + (2)^2}$$

$$= \sqrt{25+4} = \sqrt{29}$$

for $C_2 = (4, 3)$

for P_1 :

$$d = \sqrt{(5-4)^2 + (4-3)^2}$$

$$= \sqrt{1+1} = \sqrt{2}$$

$$d = \sqrt{(7-4)^2 + (6-3)^2}$$

$$= \sqrt{9+9} = \sqrt{18}$$

② edge point $P = (8, 6)$

$$C = (4, 3)$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(8-4)^2 + (6-3)^2}$$

$$= \sqrt{16+9}$$

$$= \sqrt{25}$$

③ edge point, $P = (8, 8)$

$$C = (5, 4)$$

radius,

$$r = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(8-5)^2 + (8-4)^2}$$

$$= \sqrt{9+16}$$

$$= \sqrt{25}$$

